

Detailed Cost Estimation & Budget Allocation – Project ALGOFEED

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Target Group: 25–50 low-income families (Direct social impact on food insecurity)

Fleet Size: 2–6 volunteer vehicles (Scaled for community-led, grassroots logistics)

Technological Core: Python + Google OR-Tools (Industry-standard for complex route optimization)

User Interface: Signal / Matrix Bot (Zero learning curve; high accessibility for volunteers)

Data Privacy: E2E (End-to-End) Encryption (Maximum protection for sensitive beneficiary data)

Software License: Open Source (MIT / Apache 2.0) (Ensuring long-term public utility and replicability)

Deployment Mode: Docker-based (One-click install) (Low technical barrier for other non-profits)

Target Efficiency: 15–30% mileage reduction (Improved sustainability and lower operational costs)

Technical Architecture & Logic

ALGOFEED is not a monolithic software. It consists of three distinct layers. This modular design ensures the system stays stable even when field conditions change rapidly.

1. System Components

The Optimization Core (Intelligence Layer): A Python environment using the Google OR-Tools engine. This is the „engine room.” It handles the math. It doesn’t just calculate distance; it tracks food perishability and vehicle cargo limits.

The Control Hub (Orchestration Layer): A server running in a Docker container. This is the bridge. It tracks which driver is where. It manages the database and the state of each delivery.

The Messaging Interface (Interaction Layer): Powered by Signal or Matrix protocols. We don’t build a custom app. We use existing, encrypted channels. It is secure, familiar, and lightweight.

2. Practical Workflow

The process relies on „Push-Pull” logic. The volunteer only gets the information they need at that exact moment. Data Entry: The coordinator uploads the addresses. Bakeries (pickups) and families (drop-offs).

Calculation: The VRP engine runs the model. it creates the fastest route. Chilled goods get priority.

Execution: * The driver gets a message from the bot: „Stop 1: [Name]. Navigation: [Link].” After the hand-off, the driver taps an [OK] button.

The system then „pushes” the next address immediately.

Error Handling: If a vehicle breaks down, the system sees it. It instantly redistributes the remaining stops among the other active drivers,

3.Privacy & Security

Trust is our priority.

Encryption: We use the Signal Protocol. No outside party can access the data.

Data Deletion: Beneficiary addresses are temporary. Once the delivery cycle is over, the bot wipes the history. We designed it to be GDPR-compliant by default.

Why this works for the application:

No fluff: It gets straight to the point.

Logical flow: The reviewer can follow the data from the bakery to the family.

Proven Tech: Mentioning Docker and OR-Tools shows you are using reliable, industry-standard tools.

Project Roadmap & Milestones (8 Months)

The project is divided into four distinct phases, moving from mathematical modeling to real-world social impact.

Phase 1: Architecture & Modeling (Months 1–2)

Deep Analysis: Mapping local food rescue workflows and constraint definitions.

Graph Modeling: Defining the „Road Network Graph” and delivery nodes.

Data Schema: Finalizing the database structure for volunteers and beneficiaries.

Phase 2: Core Development (Months 3–4)

VRP Engine: Implementing the Python-based routing logic (OR-Tools).

Bot Integration: Setting up the Signal/Matrix API connectors and the State Machine logic.

Coordinator Dashboard: Building the simple interface for inputting daily delivery task

Phase 3: Stress Testing & Simulation (Months 5–6)

Virtual Fleet Tests: Running simulations with 50+ addresses and 6 virtual vehicles to test algorithm stability.

Edge Case Handling: Testing the system’s response to vehicle breakdowns and „last-minute” donations.

UX Refinement: Fine-tuning the bot’s messaging flow based on internal volunteer feedback.

Phase 4: Live Pilot & Public Release (Months 7–8)

Field Deployment: 8-week live operation serving 50 families with real volunteer drivers.

Impact Assessment: Measuring mileage reduction and coordinator time-savings.

Open Source Launch: Finalizing Docker packaging and publishing the full re GitHub pository on.

Category / Milestone Description Allocation

Phase 1-2: Architecture & VRP Engine Core Python development, OR-Tools integration, and logistics modeling. €20,200

Phase 3-4: Bot & Dashboard Signal/Matrix interface, coordinator hub, and security encryption. €12,300

Phase 5-6: Field Pilot 8-week live operation, fuel subsidies for 50 families, and server hosting. €2,500

Phase 7-8: Final Release Technical documentation, Docker packaging, and GitHub finalization. €3.000

Total: € 38.000